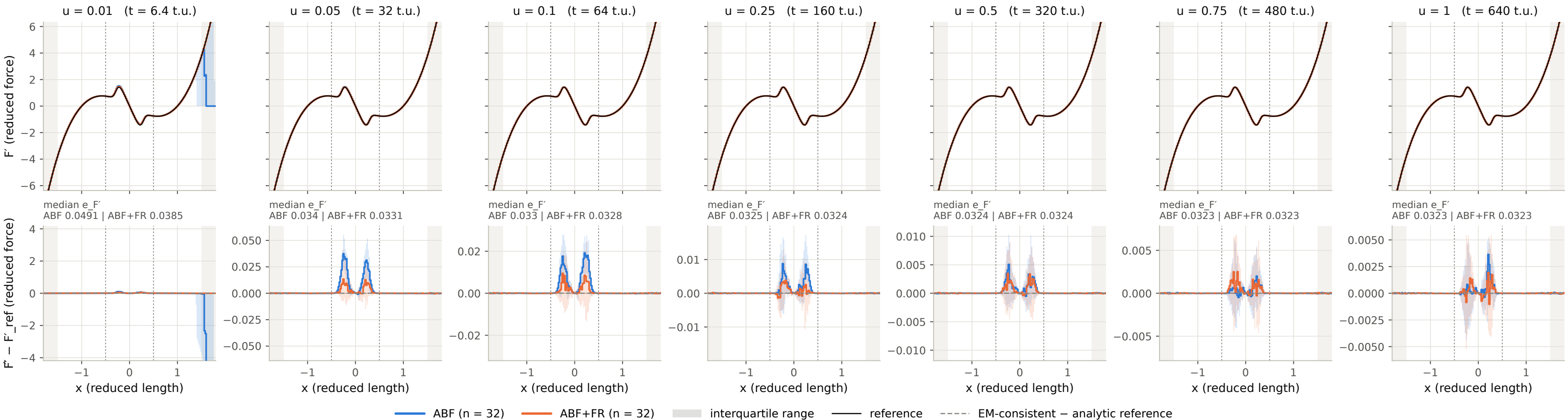


Gateway family matched_free_energy/alpha0.5 · N = 128 · T_N = 640 t.u.

n_steps = 25,600,000 · h = 2.5e-05 · B = N n_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Mean-force snapshots (bin mean force) vs reference (top) and difference (bottom)



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

Bin mean force $\Gamma = M / (C + 1)$ (step) vs the analytic F' (curve); bottom: Γ minus the bin average of F'_{ref} , whose RMS over the eval window is the floor-free companion $e_{F'_{\text{stat}}}$; $e_{F'^2} = e_{F'_{\text{stat}}}^2 + \text{floor}^2$, floor = 0.03227 (the within-bin variation of F'_{ref} : a hard lower bound of $e_{F'}$, not visible in this row). Empty bins read 0 (estimator convention).

Dashed guides: region boundaries $x = \pm 0.5$ (left well / gate / right well); shaded $|x| > 1.5$: outside the scored eval window. Top-row y range = the reference over the scored window $\pm 35\%$ of its span (larger excursions are clipped there and show in the difference row).

No smoothing.

eqb_plot_config/2 · config alpha0.5.json · 2026-10-10 14:20 UTC